Class #11 - Orthogonal Sets

1. Def: A set of vectors $\{u_1, \cdots, u_p\}$ in $\mathbf{R^n}$ is said to be an *orthogonal set* if each pair of distinct vectors from the set is orthogonal, i.e., _____whenever $i \neq j$.

2. Theorem: If $S = \{u_1, \cdots, u_p\}$ is an orthogonal set of _____vectors in $\mathbf{R^n}$, then S is linearly independent.
   (Note: If $p = n$, S is a _____for $\mathbf{R^n}$.)
   Proof:

3. Def: An *orthogonal basis* for a subspace W of $\mathbf{R^n}$ is a _____for W that is also an _____set.

4. Theorem: Let $\{u_1, \cdots, u_p\}$ be an orthogonal basis for a subspace W of $\mathbf{R^n}$. Then each $y \in W$ has a _____representation as a linear combination of $u_1, \cdots, u_p$. In fact, if

$$y = c_1 u_1 + \cdots + c_p u_p$$

then

$$c_i = \frac{y \cdot u_i}{u_i \cdot u_i} \quad (i = 1, \cdots, p).$$

Proof:

5. Example: Let $S = \{u_1, u_2, u_3\}$ where

$$u_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \quad u_2 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \quad u_3 = \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix}.$$

(a) S is an _____ _____of _____.

(b) If $y = \begin{bmatrix} 6 \\ 1 \\ -8 \end{bmatrix}$, find $[y]_S$.

6. Fix a nonzero vector $u \in \mathbf{R^n}$. (Then $W = \text{Span}\{u\}$ is a _____-dimensional _____of $\mathbf{R^n}$.) Given a vector $y \in \mathbf{R^n}$, decompose y into sum of two vectors a and b so that $a \in W$ and $b \in W^\perp$ (that is, a and b are _____).

$a =$ _____ is the *orthogonal projection of y onto* _____ (or *onto* _____), and $b =$ _____ is the *orthogonal projection of y onto* _____ (b is called the *component of y orthogonal to u*).

7. Example: Let $y = \begin{bmatrix} 7 \\ 6 \end{bmatrix}$ and $u = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$. Write y as the sum of two orthogonal vectors, one in $\text{Span}\{u\}$ and one orthogonal to u.

8. Def: A set $\{u_1, \cdots, u_p\}$ is an *orthonormal set* if it is an _____set of _____vectors.
   Note: If W =Span$\{u_1, \cdots, u_p\}$, then $\{u_1, \cdots, u_p\}$ is an _____ _____of the subspace W.

9. Example: Generate at least two distinct orthonormal bases for $\mathbf{R^3}$.